Agroforestry SmartAgroforst: Poplar biomass modelling

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Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

Keywords: Agroforestry, Short-rotation coppice, Biomass supply chain, Product cascading

1. Introduction

Agroforestry systems that integrate trees with agricultural crops (silvoarable systems) are increasingly recognised as a promising land-use option to reconcile climate mitigation, biodiversity conservation, and rural income generation on the same hectare of land (Toensmeier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such

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as poplar, willow, alder, and black locust can deliver substantial biomass yields while improving soil quality and providing multiple ecosystem services compared to conventional monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sustainable feedstock and can be used either energetically, chemically or materially (e.g. in the building sector). The latter two options are generally preferred over the energetic usage from a cleaner production perspective as biogenic carbon is stored in products during their life span which contributes to climate change mitigation.

Despite substantial research on ecological advantages and pioneering initiatives to establish AFS in practice, the share of AFS in agricultural production systems is still small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025, which accounts for about 0.02% of total agricultural land use [DEFAF]. One of the core challenges in establishing AFS systems at large scale in practice is a lack of opportunities to generate substantial revenues from woody biomasses of AFS. On the other side, the rise of bioeconomy creates substantial demands for woody biomass and requires a reliable and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low energy density, the design of cost-efficient supply chains is crucial to ensure the economic viability of AFS-based value chains. This requires coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains, which can be supported by integrated optimisation models that link stand-level growth dynamics to regional supply chain design.

In the last two decades, a large body of research has analysed biomass supply chains for bioenergy and biorefineries using mathematical programming models, often in the form of mixed-integer linear programming (MILP) (Zahraee et al., 2019; Sharma et al., 2013). These models typically optimise network configuration, feedstock sourcing, and logistics under cost or profit objectives. They often treat biomass availability as an exogenous input or rely on simplified agricultural production decisions. Moreover, downstream processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants consuming all produced biomasses. Recent work has started to integrate operational details such as various biomass types, biomass growth models and regeneration processes in multi-period supply chain models.

The present paper contributes to this literature in four ways. First, we propose a novel MILP formulation for AFS-based biomass supply chain design that explicitly links establishment decisions and consecutive harvest decisions, ensuring that harvesting decisions at each site are consistent with minimum and maximum rotation ages. Second, we embed allometric yield functions for different biomass types generating age-specific yields for different woody biomass types (like stems, branches, or leaves and twigs). Third, these different woody biomass types can be processed in specific downstream industries, like chemical or paper industry as well as energy sector. The model aims to provide strategic decision support for establishing regional AFS-based bioeconomy ecosystems connecting multiple industries and product markets.

The remainder of the paper is structured as follows. Section @ref(sec:litreview) reviews the relevant literature on agroforestry biomass systems, allometric biomass modelling, and biomass supply chain optimisation. Section @ref(sec:allometrics) introduces the allometric models adopted to represent biomass growth of different tree species and biomass types. Section @ref(sec:milp) summarises the MILP formulation for the agroforestry supply chain with age-lag constraints. Section @ref(sec:experiments) outlines the planned computational experiments for the case study and discusses expected insights for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, stand-level growth models, and biomass supply chain optimisation have each been studied extensively, but they are rarely combined in an integrated framework that links age-dependent stand dynamics to regional product-cascading value chains (Zahraee et al., 2019; Sharma et al., 2013; Espinoza et al., 2017). This section reviews (i) agroforestry systems with emphasis on species characteristics, stand management, and economics, including cascading use of biomass; (ii) biomass growth models at tree and stand level with explicit attention to component-wise partitioning into stem, branches, and residues; and (iii) biomass supply chain design using mathematical programming. The section concludes by identifying the modelling gap addressed in this paper.

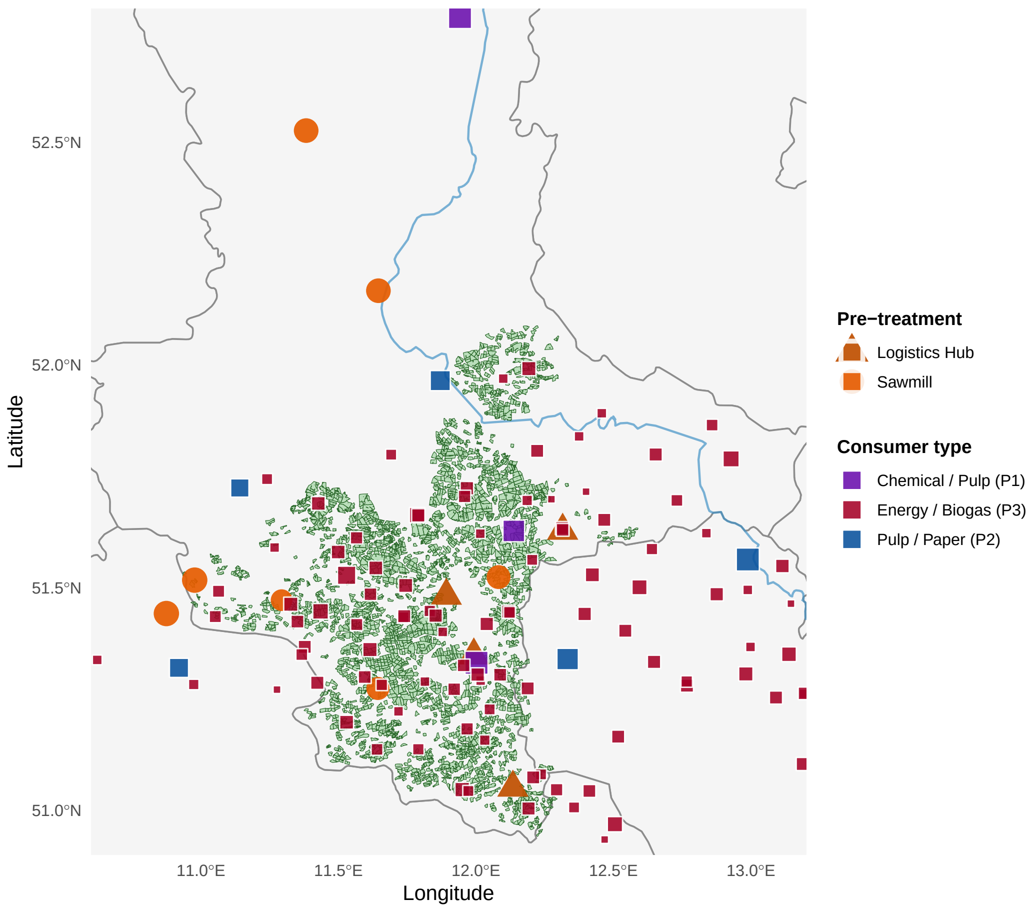

Figure 1: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Green polygons: potential KUP sites. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

2.1. Agroforestry systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance (Nair, 1993; Toensmeier, 2017). In temperate regions, short-rotation agroforestry systems (SRAFS) most commonly take the form of alley-cropping, where double or multiple tree rows alternate with crop strips, enabling simultaneous production of agricultural and woody biomass (Huber et al., 2018). Empirical trials in Southern Germany and Central Europe demonstrate that SRAFS with poplar, willow, black alder, and black locust can achieve mean annual increments (MAI) of roughly 7–10 Mg ha^{-1} yr^{-1} in aboveground woody biomass during the first rotation, with pronounced species-specific differences in growth trajectories, stand structure, and mortality (Huber et al., 2018; Trnka et al., 2008).

Economic analyses of short-rotation coppice (SRC) poplar plantations in Italy and Central Europe indicate that rotations of 5–7 years at moderate planting densities can be competitive with arable crops, but profitability is highly sensitive to yield assumptions, establishment costs, and logistics (Testa et al., 2014). Several site and management factors jointly govern biomass productivity in SRAFS and SRC systems. At the climatic level, temperature sums, solar radiation, and water availability during the growing season are the primary determinants of photosynthetic carbon gain, with drought stress particularly limiting for shallow-rooted coppice systems (Trnka et al., 2008; Huber et al., 2018). At the stand management level, planting density, pruning regime, and rotation length determine the allocation of assimilates between stem, branches, and residues, as well as the within-rotation trajectory of diameter growth, which directly conditions the proportion of merchantable stem biomass at harvest (Spinelli et al., 2011; Testa et al., 2014). At the soil level, nutrient availability and texture influence both maximum attainable biomass and growth rates, so that site-specific calibration of growth models is recommended before deployment in supply chain optimisation (Huber et al., 2016; Niemczyk et al., 2021).

From a cleaner production perspective, cascading use of biomass—prioritising high-value material applications (chemical feedstocks, pulp and paper, engineered wood) before energy recovery—is widely advocated to maximise value creation and reduce environmental burdens (Lamers et al., 2015; De Meyer et al., 2015). For agroforestry biomass, the relative shares of stem, branch, and residue fractions depend on stand age and management, so that rotation length decisions and product portfolio choices are intrinsically linked (Testa et al., 2014; Spinelli et al., 2011). MILP models can represent such cascades by defining a product quality hierarchy (e.g. chemical-grade $\succ$ pulp-grade $\succ$ energy-grade) and allowing higher-quality fractions to satisfy lower-quality demands, subject to availability and capacity constraints (De Meyer et al., 2015, 2016).

2.2. Biomass growth models

2.2.1. From tree-level allometry to stand-level yield functions

A sound stand-level biomass model for agroforestry must be traceable to measured biometric variables at the individual tree. Allometric functions of the power-law form

$$M_i = b_0 \cdot \mathrm{SBD}_i^{b_1}$$

relate the dry biomass M_i (kg) of shoot i to its stem base diameter (SBD, cm) measured 10~cm above ground, with species-specific scale parameter b_0 and exponent b_1 (Huber et al., 2016). For SRAFS species in Southern Germany, Huber et al. (2016) report a common exponent $b_1 = 2.603$ across black alder, black locust, poplar (Max~3), and willow (Inger), while b_0 ranges from 0.025 to 0.041, reflecting differences in wood density and branching architecture.

Stand-level aboveground biomass $B_{\mathrm{AG}}(t)$ (Mg ha^{-1}) at stand age t is obtained by aggregating individual shoot biomasses over all shoots per tree and multiplying by stand density $N(t)$ (trees ha^{-1}):

$$B_{\mathrm{AG}}(t) = N(t) \cdot \bar{n}_{\mathrm{shoot}}(t) \cdot \bar{M}_{\mathrm{shoot}}(t),$$

where $\bar{n}_{\mathrm{shoot}}$ is the mean number of shoots per tree and $\bar{M}_{\mathrm{shoot}}$ is the mean shoot biomass obtained by integrating the allometric function over the observed SBD distribution (Huber et al., 2018). Because SBD distributions shift and broaden over time as stands develop, this bottom-up aggregation inherently captures the age-dependence of stand biomass. In practice, where repeated diameter measurements are available, the resulting $B_{\mathrm{AG}}(t)$ trajectory can be fitted by a smooth parametric function; where only a few harvest-age data points are available, a parsimonious sigmoidal function is preferred for extrapolation (Niemczyk et al., 2021; Testa et al., 2014).

2.2.2. *Choice of model form: from Chapman–Richards to Gompertz*

The allometric functions at tree level belong to the Chapman–Richards (CR) family, which has a well-established biological derivation as the balance between anabolic resource assimilation and catabolic maintenance respiration in individual organisms (Fekedulegn et al., 1999; Zeide, 1993). The CR model is the de facto standard for individual tree diameter and height allometry in forestry because its parameters carry a direct biometric interpretation and it reduces to the von~Bertalanffy model as a special case (Fekedulegn et al., 1999).

At **stand level**, however, the argument for Chapman–Richards weakens for two reasons. First, stand-level aboveground biomass is an aggregate of hundreds of individual shoots with different diameters, subject to intra-stand competition, self-thinning mortality, and size-class redistribution over time. The resulting cumulative biomass trajectory is no longer the growth curve of a single organism but an emergent property of the population, and the biological basis of the CR formulation—anabolism versus catabolism of a single body—does not directly apply (Zeide, 1993). Second, the Chapman–Richards model is point-symmetric around its inflection, implying that roughly half the asymptotic biomass is accumulated before the inflection. Empirical data from SRC and SRAFS poplar trials consistently show that the inflection of the stand biomass trajectory occurs at a **substantially earlier** fraction of the asymptote—typically around one third—reflecting rapid initial canopy closure followed by a prolonged period of decelerating growth as competition for light intensifies (Niemczyk et al., 2021; Testa et al., 2014).

The Gompertz function

$$B(t) = A \cdot \exp\big(-\exp(-k\,(t - t_0))\big)$$

is structurally suited to this regime: its inflection point is located at $A/e \approx 0.368\,A$, well below the midpoint of the asymptote, producing the right-skewed accumulation pattern characteristic of fast-establishing coppice stands (Tjørve and Tjørve, 2017; Zeide, 1993). The Gompertz model is a limiting special case of the Richards family and has been applied successfully to model intra-annual cambial growth dynamics in broadleaved trees, where cell production over the growing season follows a Gompertz-shaped trajectory (Cuny et al., 2014). A further practical advantage is that the Gompertz helper-function decomposition used for component shares—where each fraction $f_p(t) = h_p(t)/\sum_{p'} h_{p'}(t)$

is derived from individual Gompertz sub-functions—guarantees that component fractions sum to unity at all ages by construction, without requiring additional constraints. This property simplifies both the model specification and its later integration into the MILP yield coefficient η_{pa}.

In summary, the CR model is retained at tree level because of its biological interpretability for individual-organism growth; the Gompertz model is adopted at stand level because it better captures the right-skewed biomass accumulation of coppice populations, its inflection geometry matches empirical SRC data, and its multiplicative structure supports analytically clean component partitioning.

2.2.3. Stand-level total biomass: Gompertz parameterisation

For total aboveground biomass of poplar clone Max~3 under SRAFS management, the Gompertz model is parameterised as $A = 100\ \mathrm{Mg_{DM}ha^{-1}}$, $k = 0.32\ \mathrm{yr^{-1}}$, and $t_0 = 4.0$~yr, yielding a CAI peak at approximately age~4 and a total aboveground biomass of roughly $65_{\mathrm{Mg}}\mathrm{ha}^{-1}$ at age~10—consistent with the empirical reference range of 50–$135_{\mathrm{Mg}}\mathrm{ha}^{-1}$ reported for comparable poplar systems in the literature (Huber et al., 2018; Niemczyk et al., 2021).

2.2.4. Component partitioning: stem, branches, and residues

The three components of aboveground biomass relevant for product cascading—merchantable stem, structural branches, and fine residues (leaves, twigs, bark fragments)—differ markedly in their age trajectories and market value. Empirical studies consistently show that the stem fraction rises from roughly 55% of AGB at age~4 to over 75% at age~10, while branch and residue shares decline as stands mature (Huber et al., 2018; Testa et al., 2014). This age-dependence is biologically driven by the increasing dominance of secondary stem growth (radial increment) relative to primary shoot elongation and branching in later stand development stages.

To capture these dynamics analytically while keeping the model integrable into a MILP formulation, each component share $f_p(t)$ is represented as a normalised Gompertz helper function:

$$h_p(t) = A_p \cdot \exp\!\big(-\exp(-k_p\,(t - t_{0,p}))\big), \qquad f_p(t) = \frac{h_p(t)}{\sum_{p'} h_{p'}(t)}.$$

For Max~3 the stem helper is parameterised with a high asymptote ($A_{\text{stem}} = 1.0$) and a relatively late inflection ($t_{0,\text{stem}} = 4.0$~yr), whereas the branch helper uses a lower asymptote ($A_{\text{branch}} = 0.7$) and an earlier inflection ($t_{0,\text{branch}} = 3.0$~yr), and the residue helper saturates earliest ($A_{\text{rest}} = 0.1$, $t_{0,\text{rest}} = 2.0$~yr). This parameterisation reproduces the observed shift towards stem dominance with age and is consistent with the empirical component fractions compiled across poplar agroforestry and SRC systems spanning ages 6–13 years, as illustrated in Figure~@ref(fig:max3-growth).

The allometric pathway described above provides the scientific justification for both the shape of the total biomass curve and the age-dependent component shares modelled in Figure~@ref(fig:max3-growth): the Gompertz inflection for total AGB traces back to the accelerating then decelerating SBD growth of individual shoots, and the shift in component fractions reflects the progressively larger contribution of secondary radial stem growth relative to branching as trees mature. These component-specific yield trajectories are the direct input to the yield coefficient η_{pa} in the MILP model, where $p \in \{\text{stem, branch, residue}\}$ maps to the chemical, pulp, and energy product classes, respectively.

2.3. Biomass supply chain design

Biomass supply chain design has been extensively studied using MILP at strategic, tactical, and operational levels (Zahraee et al., 2019; Sharma et al., 2013; Espinoza et al., 2017). Typical models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Eksioglu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infrastructure and operational decisions across seasonal or annual time steps (Eksioglu et al., 2009; Zahraee et al., 2019).

Among generic multi-product frameworks, OPTIMASS explicitly models product quality cascades by defining a hierarchy of biomass classes and allowing higher-quality products to satisfy lower-quality demands, while the t-OPTIMASS extension incorporates biomass growth and regeneration over multiple periods (De Meyer et al., 2015, 2016). However, in virtually all existing formulations biomass availability is treated as

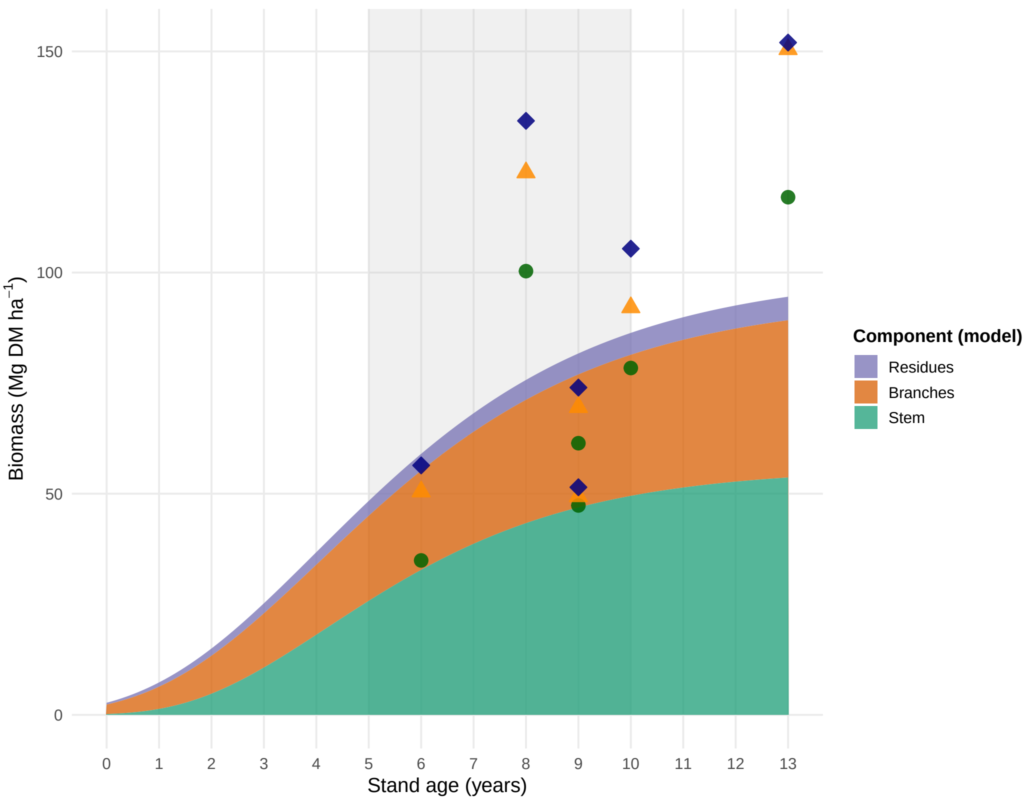

Figure 2: Modelled stand-level biomass components for poplar (Max 3) over stand age, represented as stacked Gompertz curves (shaded areas: stem green, branches orange, residues purple) and compared with empirical aboveground biomass observations from poplar agroforestry and SRC systems. Symbols denote cumulative component totals: stem (filled circles), stem+branches (triangles), total AGB (diamonds). Empirical data compiled from Jha (2018) and cited primary sources. Grey shading: typical rotation-age window (5–10 years).

an exogenous time-series input: stand age structure, rotation decisions, and coppice cycles are not represented endogenously, so that trade-offs between rotation length, stand management, and supply chain configuration cannot be explored in a unified model (De Meyer et al., 2016; Lamers et al., 2015). This gap is particularly pronounced for agroforestry systems with repeated coppice cycles, where establishment timing, consecutive harvest decisions, and age-dependent biomass quality jointly determine both the feasible supply profile and the achievable product cascade.

The MILP formulation developed in this paper addresses this gap by endogenising stand age dynamics through binary age-lag variables, coupling the component-specific yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain flow variables, and embedding a product quality cascade that allocates stem, branch, and residue fractions to chemical, pulp, and energy markets within a single optimisation model.

3. MILP model for agroforestry supply chain design

3.1. Sets, indices, and time structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \ldots, T_{\max}\}$. To represent stand establishment and harvest timing, extended time sets $\mathcal{T}^+ = \{0, \ldots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \ldots, T_{\max}\}$ are defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest. The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \ldots, T_{\max} + 1\},\ t - s \in \{A^{\min}, \ldots, A^{\max}\}\}.$$

3.2. Decision variables

Binary variables z_{ist} indicate whether, at site i, periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$. These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t) of product p at site i

in harvest period t. Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

3.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned}
\max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j\in\mathcal{J},\, p'\in\mathcal{P}:p'\le p} X_{jkp'pt} \\
& - \sum_{i} C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t\in\mathcal{T}} z_{i0t} \\
& - \sum_{i} C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t\in\mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\
& - \sum_{i,(s,t)\in\mathcal{S}:s>0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\
& - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}.
\end{aligned}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare (Testa et al., 2014). Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

3.4. Key constraints

The path connectivity constraint ensures that each harvest period at a site has exactly one predecessor and one successor in the age-lag graph, effectively forming a path over $\mathcal{T}^+$:

$$\sum_{s=0:(s,t)\in\mathcal{S}}^{t} z_{ist} = \sum_{u=t+1:(t,u)\in\mathcal{S}}^{T_{\max}+1} z_{itu} \quad \forall i\in\mathcal{I}, t\in\mathcal{T}.$$

Establishment is restricted to at most one time per site:

$$\sum_{t\in\mathcal{T}^+} z_{i0t} \le 1 \quad \forall i\in\mathcal{I}.$$

Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \mathrm{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\mathrm{harv}}.$$

Harvest quantities bound outbound flows from each site:

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\mathrm{harv}}.$$

At storage hubs, inventories obey standard balance equations with initial inventory zero:

$$S_{jpt} = S_{jp,t-1} + \sum_i X_{ijpt} - \sum_{k,p' \geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\mathrm{harv}}.$$

Storage and processing capacities are enforced via:

$$\sum_p S_{jpt} \leq \mathrm{CAP}_j^{\mathrm{stor}} \quad \forall j, t, \qquad \sum_{i,p} X_{ijpt} \leq \mathrm{CAP}_j^{\mathrm{proc}} \quad \forall j, t.$$

Demand satisfaction with cascades is modelled as described above, where higher-quality products can satisfy lower-quality demands up to maximum demand levels $D_{kpt}^{\max}$.

4. Case study and computational experiment plan

The proposed MILP model is intended to support the design of agroforestry biomass supply chains for cleaner production in specific regional contexts. In a forthcoming case study, we focus on poplar-based alley-cropping systems in a temperate European setting, informed by empirical data from SRAFS trials in Southern Germany and SRC plantations in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

We envisage the following steps and experimental scenarios for computational analysis:

1. **Parameterisation of allometric yield profiles:** Use species- and site-specific allometric models and observed growth trajectories to construct age-dependent yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative agroforestry sites. Where necessary, scenario ranges will be defined to capture uncertainty in MAI and CAI due to climate variability and management differences (Huber et al., 2016, 2018; Trnka et al., 2008).

2. **Baseline optimisation:** Solve the MILP model for a baseline configuration with fixed prices, capacities, and policy settings, obtaining optimal establishment timing, harvest schedules, and supply chain structure (locations, flows, inventories).
3. **Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ to represent alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse their impacts on profitability, product mix, and land use. This will shed light on trade-offs between shorter rotations favouring energy chips and longer rotations enabling higher shares of material products (Trnka et al., 2008; Testa et al., 2014).
4. **Product price and cascading scenarios:** Explore scenarios with different relative prices for chemical, pulp, and energy products, reflecting changes in market demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets). Assess how the product cascade hierarchy and yield quality structure influence the marginal value of species and rotation decisions (De Meyer et al., 2015; Lamers et al., 2015).
5. **Infrastructure and logistics scenarios:** Evaluate the effect of alternative storage and processing network configurations (e.g., decentralised vs. centralised hubs, different capacity levels, transport cost parameters) on the optimal spatial deployment of agroforestry sites and biomass flows (De Meyer et al., 2015, 2016; Lamers et al., 2015).
6. **Sensitivity to allometric and growth uncertainty:** Conduct parametric sensitivity and, where computationally feasible, stochastic analyses to quantify the impact of uncertainty in allometric parameters and growth responses (e.g., under drought or management differences) on optimal decisions and system robustness (Huber et al., 2018; Trnka et al., 2008).
7. **Cleaner production indicators:** For selected solutions, compute derived indicators such as biomass cascading index (share of biomass used for material products before energy), utilisation rates of higher-quality fractions, and, where coupled with LCA data, greenhouse gas emission savings relative to fossil benchmarks (Espinoza et al., 2017; Zahraee et al., 2019).

5. Conclusions

This paper outlines a research framework that couples allometric biomass growth models with a MILP-based agroforestry biomass supply chain design model featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry (Huber et al., 2018; De Meyer et al., 2015, 2016). Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

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